Algebra Solutions

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Mathematics is a game played according to certain simple rules with meaningless marks on paper.
-David Hilbert

1

Make the series of subs

$$\begin{aligned} w &= uvu^2vuv \\ &= xux^2 \qquad\qquad & v &= u^{-1}x \\ &= xy \qquad\qquad & u &= yx^{-2} \end{aligned}$$

therefore there are $|G|$ such choices since we choose x freely and must set $y = x^{-1}w$.

2

Suppose that $g \in G$ and write $A^{-1} = \{a^{-1} : a \in A\}$. Note that

$$|A| + |gA^{-1}| > |G|$$

meaning these sets must have some intersection $a_1 = ga_2^{-1}$. Then $a_1a_2 = g$ and g was arbitrary so we are done.

3

Let H_1 and H_2 have index a and b respectively. Then

$$|H_1 \cup H_2| = \frac{|G|}{a} + \frac{|G|}{b} - |H_1 \cap H_2| < |G|$$

since $a, b \geq 2$ and $|H_1 \cap H_2| \geq 1$. In the case of 3 subgroups one has

$$V_4 = \langle 1, 0\rangle \cup \langle 0, 1\rangle \cup \langle 1, 1\rangle$$

as desired.

4

Note

$$x \circ y = (x \circ x) \circ y = (y \circ x) \circ x = ((y \circ y) \circ x) \circ x = ((y \circ x) \circ y) \circ x = (y \circ x) \circ (y \circ x) = y \circ x$$

and using commutativity

$$(x \circ y) \circ z = (z \circ y) \circ x = x \circ (y \circ z)$$

5

The multiplication rule allows us to move terms between the right and the left of an equation. IE

$$a * x = b \implies (a * x) * x = b * x \implies a = b * x$$

$$x * a = b \implies x * (x * a) = x * b \implies a = x * b$$

and the reverse implications follow by symmetry. Therefore:

$$\begin{aligned} y * x = x * y &\Leftrightarrow x * (y * x) = y \\ &\Leftrightarrow x = y * (y * x) \\ &\Leftrightarrow x = x \end{aligned}$$

The counterexample for associativity is given by the "rock paper scissors" operation:

$*$	a	b	c
a	a	c	b
b	c	b	a
c	b	a	c

6

Note that

$$\begin{aligned} aba &= ba^2b \\ ab &= ba^2ba^2 && \text{right multiply by } a^2 \\ ab^2 &= ba^2ba^2b \end{aligned}$$

the RHS is symmetrical and we could have just as easily derived $b^2a = ba^2ba^2b$ by performing the same steps on the opposite side. Therefore $ab^2 = b^2a$ and all the even powers of b commute with a. In particular $b^{2n} = b$ since $b^{2n-1} = e$ so b commutes with a by (b). The first relation is then

$$aba = ba^2b \implies a^2b = a^2b^2 \implies b = e$$

as desired.

7

First, we rewrite the relation $ABA^{-1}B = 1$ as $ABA^{-1} = B^{-1}$. This indicates that conjugation by A inverts B. By extension, we can move B's past A's using the rule:

$$B^jA^i = A^iB^{j(-1)^i}$$

Every element $C \in G$ can therefore be written in the normal form $C = A^iB^j$, where $0 \leq i \leq 3$ and $0 \leq j \leq 6$.

(a) We compute the square of an arbitrary element $C = A^iB^j \in G$:

$$C^2 = (A^iB^j)^2 = A^iB^jA^iB^j = A^{2i}B^{j(-1)^i+j}$$

We evaluate this for each possible value of $i \pmod 4$:

$i = 0$**:** $C^2 = A^0B^{j+j} = B^{2j}$. Because $\gcd(2, 7) = 1$, as j ranges from 0 to 6, the powers $2j \pmod 7$ permute the elements. This generates all 7 elements of the subgroup $\langle B \rangle$.

$i = 1$**:** $C^2 = A^2B^{-j+j} = A^2B^0 = A^2$. This yields exactly 1 element, regardless of j.

$i = 2$**:** $C^2 = A^4B^{j+j} = B^{2j}$ (since $A^4 = 1$). This yields the exact same 7 elements found in the $i = 0$ case.

$i = 3$**:** $C^2 = A^6B^{-j+j} = A^2$. This yields the exact same element found in the $i = 1$ case.

Counting the distinct outcomes, there are 7 elements from the $i \in \{0, 2\}$ cases, and 1 element from the $i \in \{1, 3\}$ cases. There are 8 elements of the form C^2 in G.

(b) Based on the casework above, the distinct squares consist of all powers of B, plus the element A^2. Written as words in A and B, the 8 squares are:

$$1,\ B,\ B^2,\ B^3,\ B^4,\ B^5,\ B^6,\ A^2$$

8

We note that $a^3 = e$ since $e \in H$ and Now note that the inverse of an element has the same order as the element and we can write

$$xa = [(xa)^{-1}]^2 = (a^2x^{-1})^2 = a^2x^{-1}a^2x^{-1}$$
$$xa^2 = [(xa^2)^{-1}]^2 = (ax^{-1})^2 = ax^{-1}ax^{-1}$$

If we apply this identity to a segment of an expression we get:

$$x_1ax_2ax_3 = x_1a(a^2x_2^{-1}a^2x_2^{-1})x_3 = (x_1x_2^{-1})a^2(x_2^{-1}x_3) = x_1'a^2x_2'$$
$$x_1a^2x_2a^2x_3 = x_1a^2(ax_2^{-1}ax_2^{-1})x_3 = (x_1x_2^{-1})a(x_2^{-1}x_3) = x_1'ax_2'$$
$$x_1a^2x_2ax_3 = x_1a^2(a^2x_2^{-1}a^2x_2^{-1})x_3 = x_1ax_2^{-1}a^2(x_2^{-1}x_3) = x_1ax_2'a^2x_3'$$

We now describe an algorithm which will apply to some arbitrary element of Pa^{-1}:

$$p = x_1ax_2a\cdots x_n$$

First, we apply the first or the second equation on any adjascent occurances of a, a or a^2, a^2 to reduce it. Once we cannot do this any more, we use the first equation to "commute" occurances of a and a^2 to put all the as on the left. After this, we repeat until we cannot anymore. We will eventally reach something of one of the forms:

$$x_1ax_2 \qquad x_1a^2x_2 \qquad x_1ax_2a^2x_3$$

Therefore any element of P can be reduced to one of the forms:

$$x_1ax_2a \qquad x_1a^2x_2a \qquad x_1ax_2a^2x_3a$$

However the third expression can be transformed

$$x_1a(x_2a^2x_3ae) = x_1ax_2ax_3'a^2x_4' = x_1'a^2x_2'a^2x_4' = x_1''ax_2''$$

using the third, first, and second equation respectively. Therefore any element of P can be written in one of the forms

$$x_1ax_2a \qquad x_1a^2x_2a \qquad x_1ax_2$$

and each expression contributes at most h^2 elements.

9

(a) Note that:

$$B(c,c) = B(B(a,b), B(a,b)) = B(a,b) = c$$

(b) Note that:

$$B(c,x) = B(B(c,c), B(y,x)) = B(c, B(y,x)) = B(B(a,b), B(y,x)) = B(a,x)$$

(c) Consider the set $S = \{0,1\} \times \{0,1\}$ with:

$$B((a,b),(c,d)) = (a,d)$$

All the requirements can be checked quite easily.

10

The polynomial is irreducible exactly when the discriminant $b^2 - 4c$ is not a square. Adding k effectively allows us to make c arbitrary, so setting

$$c = 4^{-1}(b^2 - r)$$

the discriminant becomes r. Therefore there is a one-to-one mapping between the values of k and the resulting discriminant of the polynomial. Exactly $(p^k + 1)/2$ of the elements of $\mathbb{F}_{p^k}$ are squares, so our answer is

$$\frac{p^k - 1}{2}$$

since that is the number of non-squares.

11

The statement is true. Take any derangment σ of the finite set F (that is, a permutation which does not fix any element). Then define

$$x * y = \sigma(x)$$

so that any case of associativity reduces to

$$\sigma(x) = \sigma^2(x) \iff x = \sigma(x)$$

which is never true by the derangment property.

12

Let $m = |S| = (p^2 - 1)/2$ and $N = |S \cap 2S|$. Note that S and $2S$ each contain exactly one element from every pair $\{a, -a\}$ in $F^\times$. Their choices agree on N pairs and have opposite signs on the other $m - N$ pairs, so

$$\prod_{a \in S} (2a) = (-1)^{m-N} \prod_{a \in S} a$$

and canceling the nonzero product gives

$$2^m = (-1)^{m-N}$$

Now $p - 1 \mid m$ and $2^{p-1} = 1$ in F, so $2^m = 1$. Since p is odd, m is even and therefore

$$(-1)^N = 1$$

meaning N is even as desired.

13

We claim that given a and b in S we can always solve the equations

$$ax = b \qquad ya = b$$

at which point the set is a group (proof below). This is because we can let $x = a^{|a|-1}b$ and $y = ba^{|a|-1}$.

Now we need to show that G is a group if we can solve the equations above. Firstly there is a left identity since if we construct e_r, y so that $ae_r = a$ and $ya = b$ we see

$$ae_r = a \implies yae_r = ya \implies be_r = b$$

so since b was arbitrary e_r is a right identity. We can also construct a left identity e_l and note

$$e_l = e_l e_r = e_r \implies e_l = e_r$$

so there is an identity. Inverse follows from solving $ax = e$ and $ya = e$. The left and right inverses are the same since

$$a_l^{-1} = a_l^{-1} a a_r^{-1} = a_r^{-1}$$

as desired.

14

The claim is true. Consider the directed graph whose vertices are elements of G and whose edges correpond to multiplication by a or b. IE there is an edge $g \to h$ iff $h = ga$ or $h = gb$. We note that every vertex has exactly two edges going in and two edges going out. The problem asks us to find a Eulerian circuit. This is an application of the following theorem:

Theorem 0.1 (Eulerian Circuit Theorem)**.** A graph has a Eulerian Circuit (a circuit which uses every edge exactly once) iff the following conditions are satisified:

(a) The in-degree of every vertex is equal to the out-degree.

(b) All vertices with non-zero degree belong to a single strongly connected component.

Proof. The forward direction is clear. Therefore suppose that a graph satisfies the conditions. We will induct on the number of edges. Consider an algorithm which simply travels across any unused edge. We will eventually be stuck when we have used every "out" edge of a vertex, meaning we must have returned to the starting vertex. If the cycle includes every edge then we are done (note that by connectedness this is equivalent to using every edge of the vertices we have traveled on). Otherwise subtract the used edges and contruct a Eulerian circuit on the remaining edges. We can then stitch these two circuits together by starting on a vertex which they both travel on, doing one circuit and then doing the other. □

15

Note that

$$x^2 - y^2 = 1 \iff (x-y)(x+y) = 1$$

Choosing $x - y \in \mathbb{F}_p^\times$ determines $x + y$, and hence determines x, y since p is odd. Therefore there are exactly $p - 1$ ordered pairs (x, y). Let N be the number of common values $n = x^2 = y^2 + 1$. Each value contributes four pairs, except $n = 1$ contributes two, and $n = 0$ contributes two when -1 is a square. Let $\delta = 1$ if $p \equiv 1 \pmod 4$ and $\delta = 0$ otherwise. Then

$$p - 1 = 4N - 2 - 2\delta$$

so

$$N = \begin{cases} \frac{p+3}{4}, & p \equiv 1 \pmod 4 \\ \frac{p+1}{4}, & p \equiv 3 \pmod 4 \end{cases}$$

as desired.

16

Fix an element $A \in \mathcal{M}$. Note that for all $B \in \mathcal{M}$:

$$A, B \text{ commute or anticommute by (iii)} \implies A^2 \text{ and } B \text{ commute} \implies A^2 \text{ commutes with everything in } \mathcal{M}$$

In particular by property (iv) the square of every element of $\mathcal{M}$ is I. Now we need a second property. Find some $B \in \mathcal{M}$ which anticommutes with A by (iv) and compute

$$AB = -BA \implies A = -BAB^{-1} \implies \operatorname{tr}(A) = 0$$

since B is invertable (by $B^2 = I$). Now I claim that the elements of $\mathcal{M}$ are linearily independent. Suppose FTSOC that their is a linear relation

$$\sum_{M \in \mathcal{M}} c_M M = 0$$

taking trace on both sides kills all the terms except I and yields:

$$\operatorname{tr}(c_I I) = 0 \implies c_I = 0$$

Now multiply by A to get the new linear relation

$$\sum_{M \in \mathcal{M}} c_M MA = 0$$

the new term corresponding to the identity matrix is c_A since $A \in \mathcal{M}$ is the unique element which is I after multiplication by A. Therefore $c_A = 0$. Since A was arbitrary this proves linear independance, and the size of the set is bounded by n^2 since the $n \times n$ matrices have dimension n^2.

17

For $0 \leq x \leq 99$ let

$$f(x) = 101x - 100 \cdot 2^x \pmod{10100}$$

Suppose that $f(a) + f(b) = f(c) + f(d)$. By the Chinese remainder theorem

$$a + b \equiv c + d \pmod{100} \qquad 2^a + 2^b \equiv 2^c + 2^d \pmod{101}$$

Therefore in $\mathbb{F}_{101}$ the two pairs have the same sum and product:

$$2^a + 2^b = 2^c + 2^d \qquad 2^a 2^b = 2^c 2^d$$

The two pairs are roots, with multiplicity, of the same monic quadratic, so

$$\{2^a, 2^b\} = \{2^c, 2^d\}$$

Since 2 has order 100 modulo 101 and all indices lie between 0 and 99, this gives

$$\{a, b\} = \{c, d\}$$

as desired.

18

Fix an element $a \in A$, I claim that

$$c > b \iff (a, b, c) \in S$$

induces a total order on A, where we consider a as the smallest element of A.

- By (2) $(a, y, x) \in S$ iff $(a, x, y) \notin S$ which means

  $$x > y \iff y \not> x$$

  which proves antisymmetry and totality.

- To prove transitivity

  $$c > b,\ d > c \implies d > b$$

  we need to show that:

  $$(a, b, c),\ (a, c, d) \in S \implies (a, b, d) \in S$$

  However condition (3) says exactly that

  $$(a, b, c),\ (a, c, d) \in S \iff (a, b, d),\ (b, c, d) \in S$$

  after using (2) to cycle some of the triples. This is a stronger statement so transitivity is proved.

Using the total order, we define a function $g : A \to \mathbb{R}$ such that $g(x) > g(y)$ iff $x > y$. Suppose that $g(x) < g(y) < g(z)$. Then we have that

$$(a, x, y),\ (a, y, z) \in S \iff (a, x, z),\ (x, y, z) \in S$$

by condition (3). Since we have shown that $(x, y, z) \in S$ we are done. There are some special cases I have ignored where one of the x, y, z is a, but it works out in the end.

19

Compute

$$\phi(x)\phi(e)\phi(x^{-1}) = \phi(e)\phi(x)\phi(x^{-1}) \implies \phi(x)\phi(e) = \phi(e)\phi(x)$$

so $\phi(e)$ commutes with any $\phi(x)$. Now compute

$$\phi(x)\phi(y)\phi((xy)^{-1}) = \phi(xy)\phi(e)\phi((xy)^{-1}) \implies \phi(x)\phi(y) = \phi(xy)\phi(e)$$

Now, choose $a = \phi(e)^{-1}$ and define $\psi(x) = a\phi(x)$. We can now verify the homomorphism property:

$$\psi(x)\psi(y) = (a\phi(x))(a\phi(y)) = a^2\phi(x)\phi(y)$$

Substitute our identity $\phi(x)\phi(y) = \phi(xy)\phi(e)$:

$$\psi(x)\psi(y) = a^2\phi(xy)\phi(e) = a\phi(xy)(a\phi(e))$$

Since $a = \phi(e)^{-1}$, $a\phi(e)e$ and

$$\psi(x)\psi(y) = a\phi(xy) = \psi(xy)$$

20

Let the given condition be our primary axiom:

$$(x * y) * x = y \quad \text{for all } x, y \in S$$

Substitute $x = b * a$ and $y = b$ into the axiom:

$$((b * a) * b) * (b * a) = b$$

From the original axiom, we already know that $(b * a) * b = a$. Substituting that in yields

$$a * (b * a) = b$$

as desired.

21

Consider the set

$$X = \{(x_1, \dots, x_p \mid x_1) \cdots x_n = e\}$$

and note that $\mathbb{Z}/p$ acts on the set via cyclic rotation of the tuples. The fixed points are exactly the tuples of the form $(x, \dots, x)$ with $x^p = e$. Therefore $|X^G|$ is exactly the number of elements of order p.

If G has any elements of order p, then p must divide the order of G. Now note that

$$|X| = |G|^{p-1} = |X^G| + p|O|$$

where O is the set of nontrivial orbits. Taking the equation modulo p we see that

$$|X^G| \equiv 0 \pmod{p}$$

however $|X^G|$ corresponds to elements having order 0 or p. Therefore $|X^G| = n + 1$ and we are done.

22

Start with the empty sequence and let $S = \{e\}$ be its set of subsequence products. As long as $S^{-1}S \neq G$, choose $g \notin S^{-1}S$ and append it to the sequence. The new set of subsequence products is $S \cup Sg$. Note that

$$s = tg \implies g = t^{-1}s \in S^{-1}S$$

for $s, t \in S$, so $S \cap Sg = \emptyset$. Each step therefore doubles $|S|$, meaning the process must stop with $S^{-1}S = G$ after some k steps, where

$$2^k = |S| \leq |G| \implies k \leq \log_2 |G|$$

If the chosen sequence is $g_1, \dots, g_k$, now use

$$g_k^{-1}, \dots, g_1^{-1}, g_1, \dots, g_k$$

The first block gives every s^{-1} with $s \in S$ as a subsequence product, and the second gives every $t \in S$. Therefore every element of $S^{-1}S = G$ is a subsequence product of this sequence. Its length is

$$2k \leq 2 \log_2 |G| = \frac{2}{\ln 2} \ln |G|$$

so $c = 2/\ln 2$ works as desired.

23

By the structure theorem for finite abelian groups, the structure of such a group must be of the form

$$C_{2^{k_1}} \times \cdots \times C_{2^{k_n}}$$

a product of cyclic groups of order 2^{k_i}. Now consider the following. An element of G is specified elements $(g_1, \dots, g_n)$ from each factor, and because all the orders are powers of 2

$$\operatorname{ord}(g_1, \dots, g_n) = \max(\operatorname{ord}(g_1), \dots, \operatorname{ord}(g_n))$$

I know desire to compute

$$L_r = \text{number of elements of order} \leq 2^r$$

which is exactly the number of ways to choose from each factor an element of order $\leq 2^r$. Note that $C_{2^{k_i}}$ has $\min(2^{k_i}, 2^r)$ elements of order $\leq 2^r$. Therefore

$$L_r = \prod_i \min(2^{k_i}, 2^r)$$

Now note that the product of the orders of all the elements is

$$\prod_k (2^k)^{E_k} \qquad E_k := L_k - L_{k-1}$$

where E_k is the number of elements order exactly 2^k. Taking $\log_2$ we want to compute

$$\begin{aligned}\sum_k kE_k &= \sum_k k(L_k - L_{k-1}) \\ &= KL_K - \sum_{j=0}^{K-1} L_j \qquad\qquad K := \max(k_i) \\ &= K|G| - \sum_{j=0}^{K-1} L_j\end{aligned}$$

Now by setting this equal to 2009 and subtracting $L_0 = 1$ from both sides we get :

$$K|G| - \sum_{j=1}^{K-1} L_j = 2010$$

Define

$$d_j := \#\{k_i \text{ with } k_i \geq j\}$$

to be the number of factors of at least certain size. We notice that

$$L_r = \prod_i \min(2^{k_i}, 2^r) = \prod_{i=1}^{r} 2^{d_i}$$

The highest power of 2 dividing 2010 is 2^1, forcing $c_1 = d_1 = 1$ (otherwise 4 divides all terms). Thus, G has exactly one cyclic factor, making it the cyclic group C_{2^K}. This implies $L_j = 2^j$ for all j. Substituting this into our original sum yields:

$$K2^K - \sum_{j=1}^{K-1} 2^j = 2010$$

Evaluating the geometric series and writing 2009 in its binary representation, we obtain:

$$(K-1)2^K = 11111011000_2$$

The binary representation of the right-hand side has exactly three trailing zeros, which requires $K \leq 3$. However, testing the maximum possible value $K = 3$ yields $(3-1)2^3 = 16 \neq 2008$. Therefore, no such K exists.

24

Suppose FTSOC that $a, b \in G$ are such that $a \times b \neq 0$. Then

$$a \qquad a \times b \qquad a \times (a \times b)$$

are three orthogonal vectors, which we will rename v_1, v_2, v_3. We note that $-v_2 = b \times a$ is also in the group. If their magnitudes are k_1, k_2, k_3 then

$$v_1 \times v_2 = \frac{k_1 k_2}{k_3} v_3 \qquad v_1 \times v_3 = -\frac{k_1 k_3}{k_2} v_2 \qquad v_2 \times v_3 = \frac{k_2 k_3}{k_1} v_1$$

Now note that by applying the cross product formulas:

$$(v_1 v_2)v_2 = \frac{k_1 k_2}{k_3} v_3 \cdot v_2 = -k_2^2 v_1 \qquad v_1 v_2 v_2 = v_1(v_2 v_2)$$

This requires that v_2v_2 is on the same line as v_1 since otherwise a nonzero cross product with v_1 can only be orthogonal to v_1. Then by symmetry v_2v_2 is also on the same line as v_3 and is 0. On the other hand

$$(v_1(-v_2))v_2 = -\frac{k_1k_2}{k_3}v_3 \cdot v_2 = k_2^2 v_1$$

also requiring $(-v_2)v_2 = 0$ by the same argument, this is impossible since $v_2 \neq -v_2$ by assumption.

25

This is actually a linear algebra problem. Take the free $\mathbb{C}$ vector space indexed by the elements of G. each g_i can be thought of as a linear transformation of the vector space via permuting the indices by the group law. Then the matrix

$$M = \frac{1}{k}\sum_{i=1}^{k} g_i$$

describes the evolution of the probabilities. Extend $g_1, \dots, g_k$ to an enumeration of G. In particular if $v_r = (v_{1r}, \dots, v_{nr})$ where v_{ar} is the probability of landing on g_a after r multiplications then $Mv_r = v_{r+1}$ and we are asked to consider:

$$\lim_{m\to\infty} \frac{1}{b^{2m}} \sum_{i=1}^{n} ((M^m x - v)_i)^2 \qquad x = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix} \qquad v = \begin{bmatrix} 1/n \\ \vdots \\ 1/n \end{bmatrix}$$

Since the g_i generate G and $g_1 = e$, for some r the r-steps transition probability between any two elements of G must be positive. Therefore M^r is a strictly positive matrix. IE M is primitive, so by Perron Frobenius it has a simple maximal eigenvalue and all other eigenvalues have smaller magnitude. Since the eigenvector v is positive, it is the Frobenius eigenvector with eigenvalue 1. By the Perron projection we can write:

$$\lim_{m\to\infty} M^m = vw^T \qquad w^T v = 1$$

Now note that

$$M^T = \frac{1}{k}\sum_{i=1}^{k} g_i^{-1}$$

which commutes with M since the group is abelian. Therefore M is a normal matrix. Since the commuting still holds in the limit

$$[vw^T, wv^T] = 0 \implies vw^T wv^T = wv^T vw^T \implies c_1 vv^T = c_2 ww^T$$

implying that w is a scalar multiple of v. Since $w^T v = 1$ and $v^T v = 1/n$, we have $w = nv$. Since a normal matrix has orthogonal eigenvectors we can write

$$M = nvv^T + B \qquad B \text{ normal} \qquad Bvv^T = vv^T B = 0$$

Now write:

$$\sum_{i=1}^{n} ((M^m x - v)_i)^2 = \sum_{i=1}^{n} (((nvv^T + B^m)x - v)_i)^2 = \sum_{i=1}^{n} (B^m x)_i^2$$

at which point we can write $x = \sum_j c_j u_j$ in an orthonormal eigenbasis of B, with eigenvalues λ_j, and set b to be the largest $|\lambda_j|$ for which $c_j \neq 0$. Since $Bx = Mx - v \neq 0$ because $k < n$, we have $0 < b < 1$. Therefore

$$\lim_{m\to\infty} \frac{1}{b^{2m}} \sum_{i=1}^{n} (B^m x)_i^2 = \lim_{m\to\infty} \sum_j |c_j|^2 \left|\frac{\lambda_j}{b}\right|^{2m} = \sum_{|\lambda_j|=b} |c_j|^2 > 0$$

as desired.

26

I claim that there is a right identity of S. For some $a \in S$ find $e_r \in S$ with

$$ae_r = a$$

then find y with $ya = b$ and left multiply so

$$be_r = b$$

and since b is arbitrary e_r is a right identity. If we find d such that $cd = e_r$ we see that

$$ac = bc \implies acd = bcd \implies ae_r = be_r \implies a = b$$

as desired.

27

Consider the subgroup N formed by expressions of this form (without the length condition). We first must show $g, h \in N$ so that $N = G$. Note that

$$g(hg^{-1})^{|G|-1}h = ggh^{-1}h = g^2$$

is in N. Therefore since g has odd order $g \in N$ and now $h \in N$ as well.

Now we must address the length condition. Suppose we have an expression of length larger than $|G|$:

$$g^{m_1}h^{n_1}g^{m_2}h^{n_2}...$$

and write l_k to be the product of the first k terms. By pigeonhole $l_k = l_r$ for some k, r. Then cutting out all the terms in between l_k and l_r does not change the expression while reducing the length.

28

If $n = 1$ then $gh^m = e$, so $g = h^{-m}$ and the claim is clear. Otherwise write $m = qn + r$ with $0 < r < n$, since $\gcd(m, n) = 1$. We let $[x] = x - \lfloor x \rfloor$ denote the fractional part of x. Consider

$$\begin{aligned} a_k &= \left\lfloor \frac{mk}{n} \right\rfloor - \left\lfloor \frac{m(k-1)}{n} \right\rfloor \\ &= \begin{cases} q+1, & \frac{r}{n} + \left[\frac{m(k-1)}{n}\right] \geq 1 \\ q, & \text{otherwise} \end{cases} \\ &= \begin{cases} q+1, & (k-1)m \in \{-1, ..., -r\} \pmod{n} \\ q, & \text{otherwise} \end{cases} \end{aligned}$$

Note that $a_{k+n} = a_k$, so we may take indices modulo n. Since $\gcd(m, n) = 1$ we can choose $1 \leq x < n$ such that

$$xm \equiv -1 \pmod{n}$$

Now consider

$$\begin{aligned} a_{k+x} &= \begin{cases} q+1, & (k-1+x)m \in \{-1, ..., -r\} \pmod{n} \\ q, & \text{otherwise} \end{cases} \\ &= \begin{cases} q+1, & (k-1)m \in \{0, -1, ..., -r+1\} \pmod{n} \\ q, & \text{otherwise} \end{cases} \end{aligned}$$

The two residue intervals differ only at 0 and $-r$, corresponding to $k = 1$ and $k = n$. Therefore

$$a_{k+x} = \begin{cases} a_1, & k = n \\ a_n, & k = 1 \\ a_k, & \text{otherwise} \end{cases}$$

Let

$$W = gh^{a_1}gh^{a_2}\cdots gh^{a_n} = e$$

and note by conjugating the equation that a cyclic shift of W must also evaluate to e, so shifting indices by x yields

$$gh^{a_{1+x}}gh^{a_{2+x}}\cdots gh^{a_{n+x}} = e$$

Cyclically rotate both words by one block to put the common terms first, then cancel to get

$$gh^{a_n}gh^{a_1} = gh^{a_1}gh^{a_n}$$

Substitute $a_n = q + 1$ and $a_1 = q$:

$$\begin{aligned} gh^{q+1}gh^q &= gh^qgh^{q+1} \\ h^{q+1}gh^q &= h^qgh^{q+1} \end{aligned}$$

Cancel h^q on the left and right to get

$$hg = gh$$

as desired.

29

We can write power series

$$\frac{1}{1-xy} = 1 + xy + xyxy + \cdots \qquad \frac{1}{1-yx} = 1 + yx + yxyx + \cdots$$

which is unrigorous but motivates the guess

$$(1-yx)^{-1} = 1 + y(1-xy)^{-1}x$$

which can be verified using direct algebraic manipulation.

30

By our solution to problem 29 we have

$$(1-yx)^{-1} = 1 + y(1-xy)^{-1}x$$

and subbing this into the equation yields the desired identity.

31

By the fundamental theorem of finite abelian groups, G is isomorphic to the direct product of its Sylow p-subgroups. Assume, for the sake of contradiction, that a Sylow p-subgroup P of G is not cyclic. Consequently, the prime-power decomposition of P must contain at least two cyclic factors. This implies P (and therefore G) contains a subgroup isomorphic to $\mathbb{Z}_p \times \mathbb{Z}_p$. The elementary abelian group $\mathbb{Z}_p \times \mathbb{Z}_p$ has multiple distinct subgroups of order p (take $\langle 1, 0\rangle$ and $\langle 0, 1\rangle$). This contradicts the hypothesis that G possesses exactly one subgroup of order p. Thus, every Sylow p-subgroup of G must be cyclic. Since G is the direct product of cyclic groups with pairwise coprime prime-power orders, G is cyclic by Chinese remainder theorem.

32

I first claim that f almost preserves the order of real numbers. That is:

$$a \geq b \implies f(a) \geq f(b)$$

Let $a = b + r$ for $r \in \mathbb{R}^+$. Since

$$f(a) - f(b) = f(a - b) = f(r)$$

we only need to show that $f(r) \geq 0$. Since r is a positive real number it is x^2 for some x. Now

$$f(r) = f(x^2) = f(x)^2 \geq 0$$

as desired.

Now I claim that f fixes the rational numbers. Note

$$1 = f(1) = f(\underbrace{1/n + \cdots + 1/n}_{n \text{ times}}) = nf(1/n) \implies f(1/n) = 1/n$$

and note $f(a/b) = af(1/b) = a/b$ as desired.

Finally I claim that f fixes $\mathbb{R}$. Suppose for the sake of contradiction that $f(x) \neq x$. Then we can find some $\epsilon > 0$ with $f(x) \notin B_\epsilon(x)$ (ball of radius epsilon around x). By density of rational numbers we find rational $r_1, r_2 \in B_\epsilon(x)$ with $r_1 < x < r_2$. Then by order preserving property

$$r_1 \leq f(x) \leq r_2 \implies f(x) \in B_\epsilon(x)$$

a contradiction.

33

(a) Note that

$$[a, b] = aba^{-1}b^{-1} = abab = 1$$

so we have a finite dim $\mathbb{F}_2$ vector space, hence finite.

(b) Note the fundamental relation $abab = b^2a^2$ which implies

$$bab = a^2b^2a^2 \qquad b^2ab = ba^2b^2a^2$$

where the first implies the second. It suffices to show the free group

$$G = \langle x_1, \ldots, x_n : g^3 = 1 \rangle$$

is finite. We will induct on the number of generators to show that every element of can be expressed in a bounded number of terms, at which point it follows that G is finite. Consider an element $g \in G$ and keep track only of occurances of x_1 or x_1^2.

$$g = -x_1 - x_1^2 - x_1^2 - \cdots$$

where each $-$ denotes some expression of $x_2, \ldots, x_n$. We can use the second relation to "commute" adjacent occurances of x_1 and x_1^2

$$-x_1^2 - x_1 - \mapsto -x_1 - x_1^2 -$$

Therefore we can gather all the x_1 on one side and all the x_1^2 on the other.

$$g = -x_1 - \cdots - x_1 - x_1^2 - \cdots - x_1^2 -$$

at which point the first relation is used to reduce adjacent occurances of x_1, x_1 and x_2, x_2

$$-x_1^2 - x_1^2 - \mapsto -x_1 - \qquad -x_1 - x_1 - \mapsto -x_1^2 -$$

reducing the total number of occurances. We can repeat these steps until we cannot continue, reducing to an expression of one of the forms

$$- \qquad -x_1 - \qquad -x_1^2 - \qquad -x_1 - x_1^2 -$$

where in every case the minimum number of terms is bounded by $A_n \leq 2 + 3A_{n-1}$ where A_{n-1} is the bound on $n - 1$ generators by the inductive hypothesis.

34

The action of G on itself by right multiplication induces an embedding

$$G \to S_{|G|}$$

into the permutations on the elements of G. Since 2 divides $|G|$, we may find an element $x \in G$ which has order 2. The action in $S_{|G|}$ therefore looks like $2n+1$ disjoint 2 cycles, so the sign of this permutation is $(-1)^{2n+1} = -1$. In particular the composition

$$G \to S_{|G|} \overset{\text{sgn}}{\to} \{\pm 1\}$$

has full image. Therefore the kernel is an index 2 normal subgroup.

35

I first claim that if there is a two sided inverse, it is the only right inverse. Let a be a two sided inverse to x and b a right inverse to x. Then

$$a = axb = b$$

proving that there cannot be a two sided inverse. Let y be a right inverse to x and let $w = 1 - yx$ which is nonzero. Note that

$$xw = x - xyx = x - x = 0$$

so the elements

$$y_n = y + wx^n$$

are all right inverses to x. I claim that they are distinct. Suppose FTSOC that

$$\begin{aligned} y_{n_1} = y_{n_2} &\implies wx^{n_1} = wx^{n_2} \\ &\implies wx^k = w && \text{right multiply by } y^{n_2} \text{ and define } k := n_1 - n_2 \\ &\implies wx^{k-1} = wy && \text{right multiply by } y \end{aligned}$$

at this point we note that

$$wy = y - yxy = y - y = 0$$

so we have

$$wx^{k-1} = 0 \implies w = 0$$

after right multiplying by y^{k-1}. This is a contradiction as desired.

36

It suffices to write $xyz + xyz$ as a sum of squares. Note that

$$(x+yz)^2 = xyz + yzx \qquad (xy+z)^2 = xyz + zxy \qquad (y+zx)^2 = yzx + zxy$$

Therefore:

$$(x+yz)^2 + (xy+z)^2 - (y+zx)^2 = 2xyz$$

37

Note that

$$(a+b)(y+x) = (a+b)y + (a+b)x = ay + by + ax + bx$$

and

$$(a+b)(y+x) = a(y+x) + b(y+x) = ay + ax + by + bx$$

so setting these equal

$$ay + by + ax + bx = ay + ax + by + bx$$

and cancelling the leftmost and rightmost terms from both sides yields

$$by + ax = ax + by$$

as desired.

38

To solve this problem I will sequentially compute the following values:

$C_n(m)$: Let $C_n(m)$ be the number of way to pick m linearily independant vectors in $\mathbb{F}_p^n$ (order mattering). If we pick the vector sequentially, then the kth vectors can be chosen to be anything not in the span of the previous $k-1$ vectors. By linear independance there are p^{k-1} elements in this span, therefore we have:

$$C_n(m) = (p^n - 1)(p^n - p)(p^n - p^2)\cdots(p^n - p^{m-1}) = \prod_{k=0}^{m-1}(p^n - p^k)$$

I_n: Let I_n be the number of invertable $n \times n$ matrices. This is just the number of ways to pick n linearily independant columns for the matrix. Therefore:

$$I_n = C_n(n) = \prod_{k=0}^{n-1}(p^n - p^k)$$

$S_n(d)$: Let $S_n(d)$ be the number of dimension d subspaces of $\mathbb{F}_p^n$. First, we choose d linearily independant vectors spanning the subspace, and there are $C_n(d)$ way to do this. On the other hand, there are multiple choices of spanning vectors that generate the same subspace V. To account for this, note that any two basis sets for V are related uniquely by a $\dim V \times \dim V$ change of basis matrix. Therefore divide out by I_d to obtain:

$$S_n(d) = \frac{C_n(d)}{I_d} = \frac{\prod_{k=0}^{d-1}(p^n - p^k)}{\prod_{k=0}^{d-1}(p^d - p^k)}$$

$R_n(r)$: Let $R_n(r)$ be the number of $n \times n$ matrices of rank r. To pick a matrix M in this set, we need to choose the kernel space, and there are $S_n(n-r)$ ways to do this. If we extend a basis of $\ker M$ to a basis of $\mathbb{F}_p^n$ via $v_1, \ldots, v_r$ then M is uniquely determined by the image of the v_i. However the images must be linearily independant so that the rank of M is indeed r. Therefore there are $C_n(r)$ choices for the images. This gives:

$$R_n(r) = S_n(n-r) \cdot C_n(r) = \frac{\prod_{k=0}^{n-r-1}(p^n - p^k)}{\prod_{k=0}^{n-r-1}(p^{n-r} - p^k)} \prod_{k=0}^{r-1}(p^n - p^k)$$

39

(a) Consider an element $c = (c_1, c_2, ...)$. Let

$$c_i^{(k)} = \begin{cases} 0 & \text{if } i < k \\ c_i & \text{if } i \geq k \end{cases}$$

be the same sequence with the first $k-1$ terms set to 0. Note that $f(c - c^{(k)}) = 0$ since it is a finite sum, therefore

$$f(c) = f(c - c^{(k)} + c^{(k)}) = f(c - c^{(k)}) + f(c^{(k)}) = f(c^{(k)})$$

so f does not care about any finite amount of initial terms.

We call a sequence $d = (d_1, d_2, ...)$ *p-increasing* if p^k divides d_k for all k. Since every term of $d^{(k)}$ is divisible by p^k we can write

$$f(d) = f(d^{(k)}) = p^k f(d^{(k)}/p^k)$$

implying that $f(d)$ is divisible by p^k. Since k was arbitrary, this is only possible if $f(d) = 0$, so f kills every p-increasing sequence.

Now we want to show that $f(c) = 0$. By bezouts theorem we can find x_i, y_i for every i such that

$$2^i x_i + 3^i y_i = c_i$$

so define the sequences $a = (2x_1, 2^2 x_2, ...)$ and $b = (3y_1, 3^2 y_2, ...)$. By construction we have

$$f(c) = f(a + b) = f(a) + f(b)$$

however a, b are 2-increasing and 3-increasing respectively, therefore they are killed by f. This gives $f(c) = 0$ as desired.

(b) Suppose for the sake of contradiction that there are infinitely many indices $i_1 < i_2 < ...$ such that $f(e_{i_k}) \neq 0$. We will construct a sequence $c = (c_1, c_2, ...)$ which yields a contradiction via the binary representation of $f(c)$. Define $c_{i_k} = 2^{m_k}$ and $c_j = 0$ for $j \notin \{i_1, i_2, ...\}$. We choose the exponents m_k recursively to grow fast enough to prevent binary overlap:

$$m_1 = 1, \qquad m_{k+1} = m_k + \lceil \log_2 |f(e_{i_k})| \rceil + 2$$

Since $c \in G$, its evaluation $f(c)$ must equal some finite integer N. For any n, we can split c into a finite prefix and an infinite tail:

$$c = \sum_{k=1}^{n} 2^{m_k} e_{i_k} + 2^{m_{n+1}} d^{(n)}$$

where $d^{(n)}$ is an integer sequence (since all subsequent terms have at least a factor of $2^{m_{n+1}}$). Applying f to both sides, we get:

$$N = f(c) = \sum_{k=1}^{n} 2^{m_k} f(e_{i_k}) + 2^{m_{n+1}} f(d^{(n)})$$

This implies the congruence:

$$N \equiv \sum_{k=1}^{n} 2^{m_k} f(e_{i_k}) \pmod{2^{m_{n+1}}}$$

Notice our choice of m_{k+1} ensures that the magnitude of $2^{m_k} f(e_{i_k})$ is strictly less than $2^{m_{k+1}-1}$. Because the blocks of bits for each term are separated by zeros, each $f(e_{i_k}) \neq 0$ forces at least one non-zero bit into the 2-adic representation of N between the positions m_k and m_{k+1}.

Since there are infinitely many such k, the binary representation of N must contain infinitely many non-zero bits (or infinitely many alternating bit flips if $N < 0$). However, N is a fixed, finite integer, meaning its binary representation must eventually become all 0s (if $N \geq 0$) or all 1s (if $N < 0$). This is a contradiction. Therefore, $f(e_i) = 0$ for all but finitely many i.

40

We must compute the sum:

$$\sum_{g\in G}\sum_{\substack{x\in G\\ xg=gx}} 1 = \sum_{g\in G}|C_G(g)|$$

Now note that the map

$$G\times G\to G \qquad u\times v\mapsto uvu^{-1}$$

by conjugation defines a G action on itself, and we are tasked with finding the sum

$$\sum_{g\in G}|X^g|$$

where X^g are the fixed points of G under conjugation by g. This is because commuting is equivalent to being fixed by the conjugation action. Then by Burnsides lemma this is equal to

$$|G||X/G|$$

where X/G is the number of orbits under this action. IE the number of conjugacy classes. The answer is therefore:

$$|G|k(G)$$

41

This turned out to be a difficult problem about S_n reps. We work in the group algebra $\mathbb{C}[G]$. Let

$$K=\sum_{c\in\{n\text{-cycles}\}} c$$

be the conjugacy class sum of the n-cycles. We need to compute the coefficient in front of the identity in K^3. This can be determined from the trace in the regular representation so we use Fourier analysis and sum over the characters:

$$\operatorname{Tr}(K^3)=\sum_{V\in\operatorname{Irr}S_n} d_V \operatorname{tr}K^3|_V \qquad d_V=\dim V$$

We note that the sum over a coneugacy class C is scaling by $|C|\chi_V(C)/d_V$ via standard rep theory. Therefore the trace of K^3 is $\frac{|C|^3\chi_V(C)^3}{d_V^2}$. Therefore we must calculate:

$$\sum_{V\in\operatorname{Irr}S_n}\frac{|C|^3\chi_V(C)^3}{d_V}$$

The difficulty now is computing the character. The standard method is the *Frobenius formula*:

Theorem 0.2. Let $\lambda=(\lambda_1,\lambda_2,\dots,\lambda_k)\vdash n$ index an irreducible representation of S_n, and let $\mu=(\mu_1,\mu_2,\dots,\mu_m)$ denote the cycle type of an element in S_n.
Define the strictly decreasing sequence $l_i=\lambda_i+k-i$. The character $\chi^\lambda(\mu)$ is given by the coefficient of the monomial $x_1^{l_1}x_2^{l_2}\dots x_k^{l_k}$ in the expansion of the Vandermonde determinant multiplied by the power sum symmetric polynomials:

$$\chi^\lambda(\mu)=\left[x_1^{l_1}x_2^{l_2}\cdots x_k^{l_k}\right]\left(\prod_{1\le i<j\le k}(x_i-x_j)\prod_{r=1}^{\ell(\mu)}p_{\mu_r}(x_1,\dots,x_k)\right) \tag{1}$$

Since μ is an n-cycle, we can plug in $x_1^n+\cdots+x_k^n$ in place of the power sum product. The standard expansion of the Vandermonde polynomial is $\sum_{\sigma\in S_k}\operatorname{sgn}(\sigma)x_{\sigma(1)}^{k-1}x_{\sigma(2)}^{k-2}\cdots x_{\sigma(k)}^0$, and our RHS is:

$$(x_1^n+\cdots+x_k^n)\sum_{\sigma\in S_k}\operatorname{sgn}(\sigma)x_{\sigma(1)}^{k-1}x_{\sigma(2)}^{k-2}\cdots x_{\sigma(k)}^0$$

Note that $l_1 \geq k$. To match the LHS, we consider only the x_1^n term of the power sum because we need a monomial with a high enough power of x_1. This means the Vandermonde term must provide exactly power 0 for x_1, and the remaining variables $x_2, \ldots, x_k$ must take the remaining available Vandermonde powers, which are strictly decreasing. The only monomial in the Vandermonde expansion matching this requirement is:

$$\operatorname{sgn}(\sigma) x_1^0 x_2^{k-1} x_3^{k-2} \cdots x_k^1$$

Multiplied by x_1^n, this gives the term $\operatorname{sgn}(\sigma) x_1^n x_2^{k-1} \cdots x_k^1$. Matching exponents to $x_1^{l_1} x_2^{l_2} \ldots x_k^{l_k}$, we see that $l_k = \lambda_k = 1$ and working backwards we get that $\lambda_i = 1$ for all $i > 1$. Therefore the shape of λ is a hook, i.e., $\lambda = (n-c, 1^c)$ where $c = k-1$. The Vandermonde monomial $x_1^0 x_2^{k-1} \cdots x_k^1$ corresponds to shifting the standard exponent sequence $(k-1, k-2, \ldots, 0)$ to $(0, k-1, \ldots, 1)$. Moving the 0 from the end to the front requires shifting it past exactly $k-1$ elements. This takes $k-1$ adjacent transpositions, meaning the sign is:

$$(-1)^{k-1}$$

Since $c = k-1$, this evaluates to $(-1)^c$. Therefore, the character of an S_n irreducible representation evaluated on an n-cycle is:

$$\chi^\lambda((n)) = \begin{cases} (-1)^c & \text{if } \lambda = (n-c, 1^c) \\ 0 & \text{otherwise} \end{cases}$$

Going back to our trace computation:

$$\begin{aligned} \sum_{V \in \operatorname{Irr} S_n} \frac{|C|^3 \chi_V(C)^3}{d_V} &= \sum_{c=0}^{n-1} \frac{|C|^3 \chi^{(n-c,1^c)}(C)^3}{d_{(n-c,1^c)}} \\ &= (n-1)!^3 \sum_{c=0}^{n-1} \frac{(-1)^c}{\binom{n-1}{c}} \end{aligned}$$

For $0 \leq k \leq N$ note that

$$\frac{(-1)^k}{\binom{N}{k}} = \frac{N+1}{N+2} \left(\frac{(-1)^k}{\binom{N+1}{k}} - \frac{(-1)^{k+1}}{\binom{N+1}{k+1}} \right)$$

so summing from $k = 0$ to N telescopes to

$$\sum_{k=0}^{N} \frac{(-1)^k}{\binom{N}{k}} = \frac{N+1}{N+2} \left(1 + (-1)^N\right)$$

Taking $N = n-1$ and noting that n is odd gives:

$$\operatorname{Tr}(K^3) = (n-1)!^3 \sum_{c=0}^{n-1} \frac{(-1)^c}{\binom{n-1}{c}} = (n-1)!^3 \frac{2n}{n+1}$$

Now the trace is $n!$ times the coefficient in front of the identity in the regular rep (what we want to compute). So our answer is:

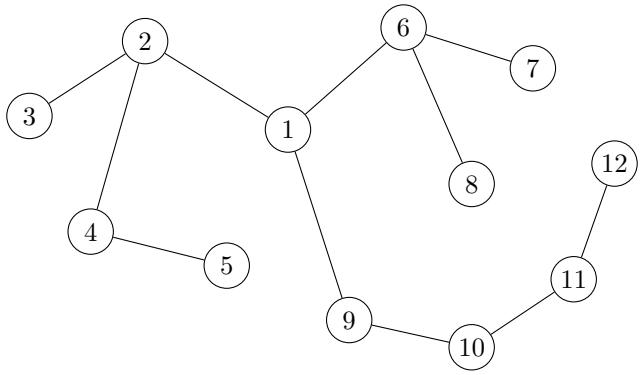
$$\operatorname{Tr}(K^3)/n! = \frac{2(n-1)!^2}{(n+1)}$$

as desired.

42

We want to think of having a tranposition (i, j) in our expression as drawing an edge between i and j in some graph of n nodes. It is clear our graph must be connected, since otherwise the cycle type of our final

expression would be decomposible. Since there are $n-1$ edges our graph must be acyclic (deleting an edge from the cycle would yield a connected graph of $n-2$ edges on n vertices, impossible).



Now we use the following lemma to construct a bridge back from this tree picture to our transpositions.

Lemma 0.3. The edges of any labeled tree, multiplied as transpositions in any order, give an n-cycle.

Proof. By induction on n, as we add edges in order the cycles of the resulting permutation are exactly the connected components of the graph. When we add the $n-1$th edge, we are connecting two cycles, combining them into an n-cycle. □

Now we have a bijection:

$$(\text{Trees on } \{1,\dots,n\}) \times (\text{edge orderings}) \longleftrightarrow \text{ways to write an } n\text{-cycle as a product of } n-1 \text{ transpositions}$$

Therefore the number of ways to write an n-cycle as a product of $n-1$ transpositions is

$$n^{n-2} \cdot (n-1)!$$

by Cayley's formula on the number of trees on labeled vertices. We only about the n-cycle $(1,2,\dots,n)$. By the symmetry of the problem we can divide by the number of n-cycles (which is $(n-1)!$) to yield

$$n^{n-2}$$

as our final answer.

43

I will generally write permutations as

$$\sigma \equiv \sigma^{-1}(1),\dots,\sigma^{-1}(n)$$

so that postcomposing with s_i swaps the entries in positions $i, i+1$.

Let R be the reflected permutation and $p = \binom{n}{2}$. Every expression

$$R = s_{i_1} \cdots s_{i_p}$$

is a reduced word, meaning that no shorter expression represents the same permutation.

We construct a bijection with standard Young tableaux of staircase shape

$$\delta = (n-1, n-2, \dots, 1)$$

using an insertion procedure.

Start with two empty tableaux P, Q. Read $i_1,\dots,i_p$ from left to right. At step t, insert $x = i_t$ into the first row of P as follows:

(a) If x is larger than every entry in the row, append x and stop

(b) Otherwise let y be the smallest entry greater than x. Replace y by x and insert y into the next row

(c) The exception to the second rule is when the row contains both x and $x+1$. In this case leave the row unchanged and insert $x+1$ into the next row

Repeat until a new box is created. Add a box in the same position in Q, labeled t.

Thus P records the inserted indices after bumping, while Q records when each box is created.

For example

$$121 \longmapsto \left(\begin{array}{cc} 1 & 2 \\ 2 & \end{array} , \begin{array}{cc} 1 & 2 \\ 3 & \end{array} \right) \qquad 212 \longmapsto \left(\begin{array}{cc} 1 & 2 \\ 2 & \end{array} , \begin{array}{cc} 1 & 3 \\ 2 & \end{array} \right)$$

We need to prove that this procedure gives a bijection. For a tableau T with strictly increasing rows and columns, let $\rho(T)$ be the word obtained by reading its rows from bottom to top, and each row from left to right.

Lemma 0.4. Suppose $\rho(T)$ is reduced. If $\rho(T)x$ is reduced, inserting x produces another such tableau T' with

$$\rho(T') \sim \rho(T)x$$

where $\sim$ denotes equivalence under the commuting and braid relations

$$s_i s_j = s_j s_i \quad (|i-j|>1) \qquad s_i s_{i+1} s_i = s_{i+1} s_i s_{i+1}$$

Conversely, every removable corner of T determines a unique inverse insertion producing T' and x with

$$\rho(T) \sim \rho(T')x$$

Proof. First consider insertion into a single increasing row U.

If x is not already in U and bumps y, write $U = AyB$, where every entry of A is smaller than x and every entry of B is larger than y. By commuting distant transpositions,

$$AyBx \sim AyxB \sim yAxB$$

which is precisely

$$Ux \sim yU'$$

where $U' = AxB$ is the new row.

If $x, x+1$ both occur, write $U = Ax(x+1)B$. The braid relation gives

$$Ax(x+1)Bx \sim Ax(x+1)xB \sim A(x+1)x(x+1)B \sim (x+1)Ax(x+1)B$$

so again

$$Ux \sim (x+1)U$$

as required by the exceptional rule.

Finally, if x occurs in U but $x+1$ does not, then writing $U = AxB$ gives

$$Ux = AxBx \sim AxxB$$

which is not reduced. Thus this case cannot occur during insertion into a reduced word.

Applying these identities successively to the rows, we see that the row-reading word of the new tableau is equivalent to the original word followed by x. In particular it remains reduced.

The rows remain strictly increasing. The columns also remain strictly increasing: the columns in which bumping occurs weakly decrease as we move downward, since an entry immediately below a bumped entry is larger. Consequently a newly inserted entry is larger than the entry above it. The only possible exception is when two consecutive bumps occur in the same column, the first being exceptional. But if $x+1$ is bumped exceptionally from column c, the row below has its entry in column $c-1$ at least $x+1$. Hence an

ordinary insertion of $x+1$ cannot occur in column c; another exceptional bump must occur instead. Thus column-strictness and the Young diagram shape are preserved.

Now we describe the inverse explicitly. Choose any removable corner, delete its entry z, and move upward through the rows. In each row U, let y be its largest entry smaller than z. There are two cases:

(a) If z does not occur in U, replace y by z

(b) If z occurs in U, leave the row unchanged and set $y = z-1$

Continue into the preceding row with $z = y$. The value leaving the first row is the recovered letter x.

We verify that this always works. At each stage the current value z came from some column c of the row below. The entry immediately above that position is smaller than z, so y exists. Moreover, if z occurs in U without $z-1$, then

$$zU = zAzB \sim AzzB$$

since every entry of A is at most $z-2$. This contradicts reducedness, so the second case is always legitimate.

Replacing y by z preserves the increasing rows. It also preserves increasing columns: if y occupies column d, then $d \geq c$, so every entry below the new z is larger than z. The only possible equality occurs when the preceding reverse bump was exceptional and $d = c$. But then the row below contains $z, z+1$ in columns $c, c+1$, forcing the current row to contain z in column $c+1$. Thus this step is exceptional as well and no replacement occurs.

Finally, each reverse step is exactly one of the preceding row identities applied backward. Therefore reducedness is preserved throughout, and the forward and reverse procedures undo each other. □

We now apply the lemma to our reduced word for R.

Every insertion adds exactly one box, and all entries of P belong to $\{1, \dots, n-1\}$. Since P is strictly increasing along rows and columns,

$$r + c - 1 \leq P_{r,c} \leq n-1$$

so every box satisfies $r + c \leq n$. Therefore the shape of P is contained in δ.

But there are exactly $p = |\delta|$ insertions, so P must fill the entire staircase.

Furthermore

$$r + c - 1 \leq P_{r,c} \leq P_{r,n-r} - (n - r - c) \leq r + c - 1$$

which forces

$$P_{r,c} = r + c - 1$$

Thus the final insertion tableau is always the same tableau

$$P = \begin{array}{ccccccc} 1 & 2 & 3 & \cdots & n-1 \\ 2 & 3 & \cdots & n-1 \\ 3 & \cdots & n-1 \\ \vdots & \cdots \\ n-1 \end{array}$$

All the information about the reduced word is therefore stored in Q.

Conversely, start with this fixed P and any standard Young tableau Q of shape δ. Note that

$$\rho(P) = (n-1)(n-2, n-1)\cdots(1, 2, \dots, n-1)$$

is a reduced word for R, as can be checked by successively moving the largest remaining element leftward.

Remove the boxes of Q in the order $p, p-1, \dots, 1$, applying reverse insertion to P each time. By the lemma every step is valid and recovers one swap index. Since the row-reading word represents R throughout, the recovered word is a reduced expression for R.

The two procedures are inverse, so we have established the bijection

$$\{\text{reduced words for } R\} \longleftrightarrow \{\text{standard Young tableaux of shape } \delta\}$$

Finally, given Q, define a_t to be the row containing t. Since row r has $n-r$ boxes, there are exactly $n-r$ occurrences of r. Moreover, the boxes labeled $1, \dots, k$ form a Young diagram, giving

$$\#\{t \le k : a_t = r+1\} \le \#\{t \le k : a_t = r\}$$

Conversely, any sequence satisfying these conditions determines a standard Young tableau by placing t in the next available box of row a_t. The prefix inequalities guarantee that the result is standard.

This establishes the desired bijection. This problem is highly connected to the theory of Coxeter groups.

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Let R be the reflected permutation, $p = \binom{n}{2}$, and

$$S = \sum_{s_{i_1} \cdots s_{i_p} = R} i_1 \cdots i_p$$

We want operators that make nonreduced words vanish and give the same result for all reduced words representing a given permutation. The *divided differences* on $\mathbb{C}[x_1, \dots, x_n]$ do exactly this:

$$\partial_i g = \frac{s_i g - g}{x_i - x_{i+1}}$$

where s_i swaps x_i and x_{i+1}. They combine an adjacent swap with a drop in polynomial degree, and satisfy

$$\partial_i^2 = 0 \qquad \partial_i \partial_j = \partial_j \partial_i \quad (|i-j| > 1) \qquad \partial_i \partial_{i+1} \partial_i = \partial_{i+1} \partial_i \partial_{i+1}$$

By the word property for adjacent transpositions, a nonreduced word therefore gives zero, while a reduced word gives an operator depending only on its permutation. Denote this operator by ∂_w.

Now set

$$D = \sum_{i=1}^{n-1} i \partial_i$$

Expanding D^p produces the weights we want. Since R is the only permutation of length p, all surviving terms represent R, giving

$$D^p = S \partial_R$$

The coefficients i now suggest a polynomial to try: differentiating x_{i+1}^i produces exactly the factor i. So choose

$$f = x_2 x_3^2 \cdots x_n^{n-1}$$

Its degree is p, as needed for the derivative interpretation. Also, when neighboring exponents differ by one, taking a divided difference just removes one factor, giving

$$\partial_i f = \frac{f}{x_{i+1}}$$

Using the standard reduced expression for R,

$$\partial_R f = \partial_{n-1}(\partial_{n-2}\partial_{n-1}) \cdots (\partial_1 \cdots \partial_{n-1}) f = 1$$

since, applying the blocks from right to left, each block lowers every positive exponent by one. Thus $S = D^p f$.

Now the weights become the coefficients of ordinary differentiation:

$$Df = \sum_{i=1}^{n-1} \frac{if}{x_{i+1}} = Ef \qquad \text{where} \qquad E = \sum_{j=1}^{n} \frac{\partial}{\partial x_j}$$

To see why this agreement persists under iteration, think of E as differentiating a simultaneous translation of all variables. A common shift $x_j \mapsto x_j + t$ commutes with every swap and leaves every difference $x_i - x_{i+1}$ unchanged. Hence E commutes with every ∂_i, so $DE = ED$ and $D^k f = E^k f$ by induction. Therefore

$$S = D^p f = E^p f = \frac{d^p}{dt^p} \prod_{j=2}^{n} (x_j + t)^{j-1} \Bigg|_{t=0} = p!$$

because the product is monic of degree p in t.